Library Management System (WEB-LMS)

Project Report

Submitted By:

Name: Alex Verma

Course: B.Tech CSE

Intern ID: RL26070107

Domain: Web Development

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1. Project Objective

The objective of the Library Management System (WEB-LMS) is to design and develop a fully functional, browser-based application that simplifies library operations. The system aims to provide an intuitive interface for managing books, members, and transactions without requiring any server-side infrastructure or database setup.

Key goals include enabling role-based access (Admin and Member), real-time tracking of book availability, automated overdue detection, and data-driven insights through interactive charts and reports. The system is built to be lightweight, responsive, and accessible across all modern devices.

2. Technologies Used

Category	Technology	Purpose
Markup	HTML5	Semantic page structure and content
Styling	CSS3	Layout, theme, and responsive design with custom properties
Logic	Vanilla JavaScript (ES6+)	All application logic without frameworks or bundlers
Icons	Lucide (CDN)	Lightweight SVG icon library for UI elements
Charts	Chart.js (CDN)	Interactive data visualizations on the Reports page
Font	Outfit (Google Fonts)	Modern sans-serif typeface
Storage	Web LocalStorage	Client-side data persistence (no database required)
Dev Server	http-server (Node.js)	Zero-config local development server

3. Design Overview

The application follows a multi-page architecture with a shared sidebar navigation and responsive layout. Key design principles include:

- **Clean Interface:** Minimal design with focus on readability and ease of use
- **Role-Based UI:** Admin and Member views with contextual menus and actions
- **Theme Support:** Light and dark mode toggle with CSS custom properties (HSL tokens)
- **Responsive Layout:** Flexbox and CSS Grid ensure usability across desktop and mobile
- **Component-Based CSS:** Reusable component styles in `components.css` (cards, tables, modals, forms, toasts)
- **Data Flow:** Centralized state management in `core.js` with getter/setter pattern over LocalStorage

4. Features Implemented

Dashboard

- KPI stat cards: Total Titles, Available Copies, Active Members, Overdue Books
- Recent Operations table with latest transaction history
- Role-based Quick Actions panel for common tasks
- Welcome banner with contextual messaging and navigation shortcuts
- Automatic overdue transaction detection on page load

Books Catalogue

- Search by title, author, or ISBN with real-time filtering
- Category filter dropdown and availability toggle
- Visual book cards with cover images, metadata, and stock indicators
- Admin: Add, edit, and delete books via modal forms with validation
- Member: One-click borrow request from the catalogue
- Out-of-stock badges for unavailable titles

Members Management (Admin Only)

- Search by name, email, or phone with instant results
- Role filter (Admin / Member) for targeted views
- Member profile cards with avatar, contact info, and active borrow count
- Full CRUD: Add, edit, and delete member profiles
- Automatic access restriction and redirect for non-Admin users

Transactions

- Issue form with autocomplete for member and book selection
- Return processing with date tracking and overdue detection
- Renewal option extending due date by 14 days
- Searchable transaction history filtered by status (Issued, Returned, Overdue)
- Color-coded status badges for quick visual identification

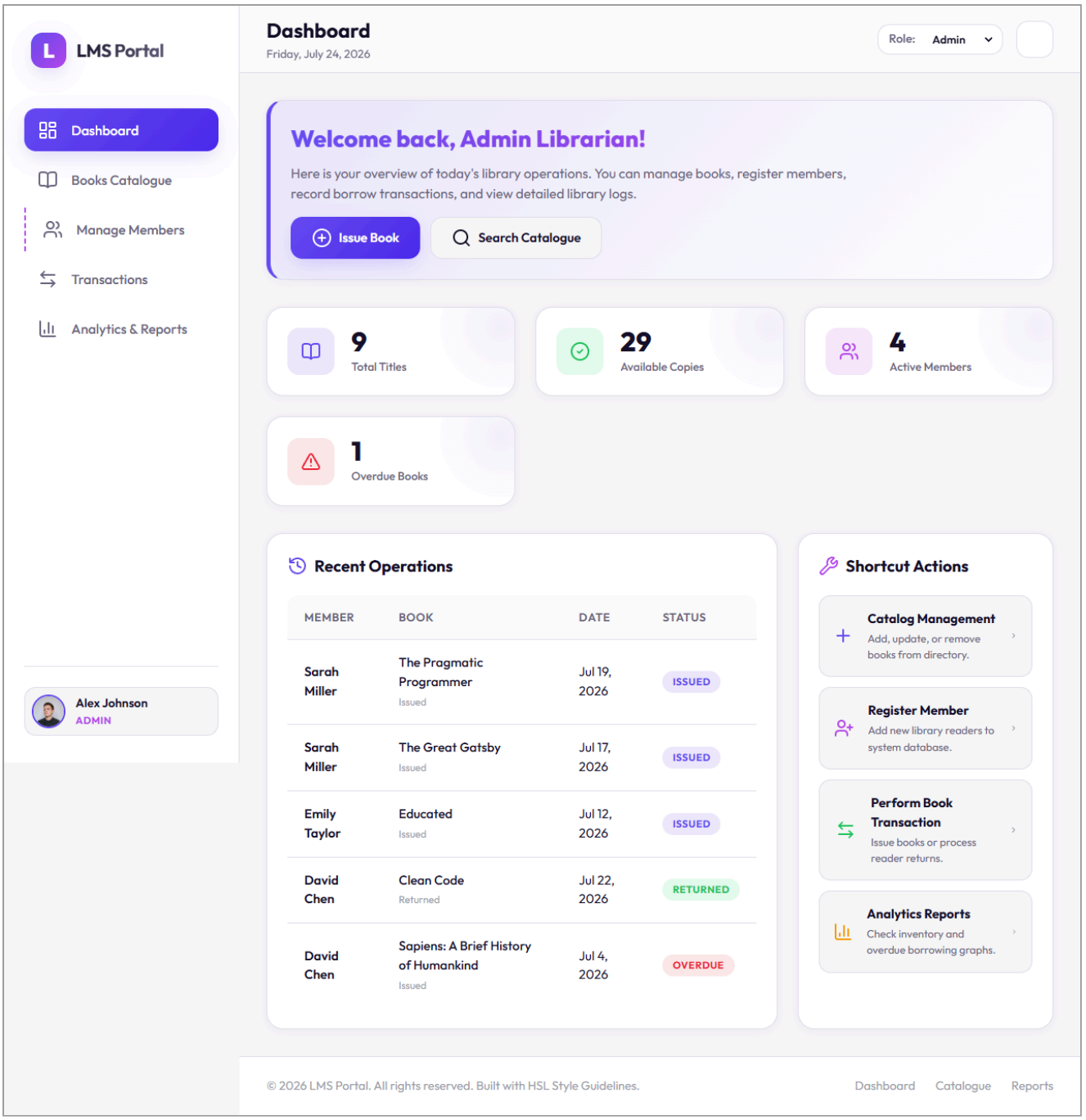
Analytics & Reports

- KPI cards: Circulation Rate, Average Loan Period, Top Genre
- Category distribution doughnut chart
- Book availability pie chart (Available vs Issued)
- Top 5 most borrowed books horizontal bar chart
- Overdue Register table with mock notice generation
- Member Insights panel showing borrowing activity per member
- Theme-aware chart redraw on dark/light mode toggle

5. Project File Structure

```
WEB-LMS/
|-- index.html              (Dashboard)
|-- books.html              (Books Catalogue)
|-- members.html            (Manage Members)
|-- transactions.html       (Transactions)
|-- reports.html            (Analytics & Reports)
|-- package.json            (npm config)
|
|-- css/
|   |-- styles.css          (Core layout & theme)
|   |-- components.css      (Reusable UI components)
|
|-- js/
|   |-- core.js             (Shared logic, storage, seed data)
|   |-- dashboard.js        (Dashboard controller)
|   |-- books.js            (Books catalogue controller)
|   |-- members.js          (Members management controller)
|   |-- transactions.js     (Transactions controller)
|   |-- reports.js          (Reports & charts controller)
```

6. Screenshots



Dashboard Page

Library Catalogue

Search, filter, and manage library books

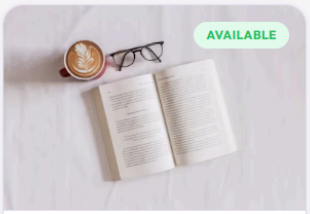
Role: Admin

Search by title, author, or ISBN...

All Categories

All Availability

+ Add New Book



COMPUTERS

Clean Code

by Robert C. Martin

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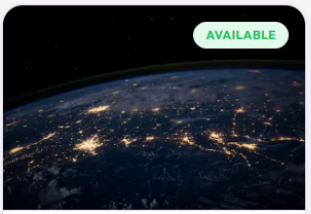
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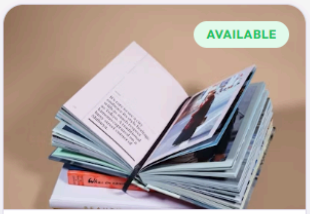
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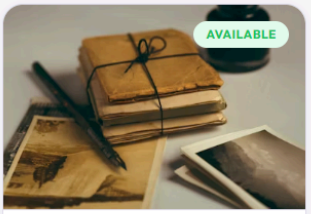
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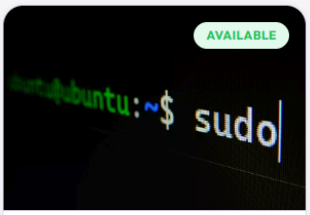
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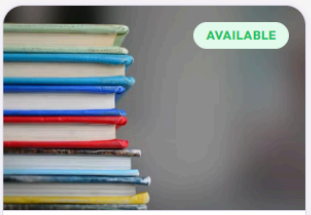
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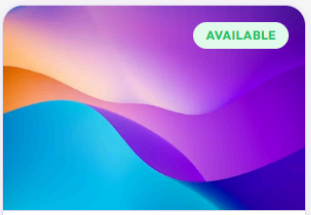
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BUSINESS

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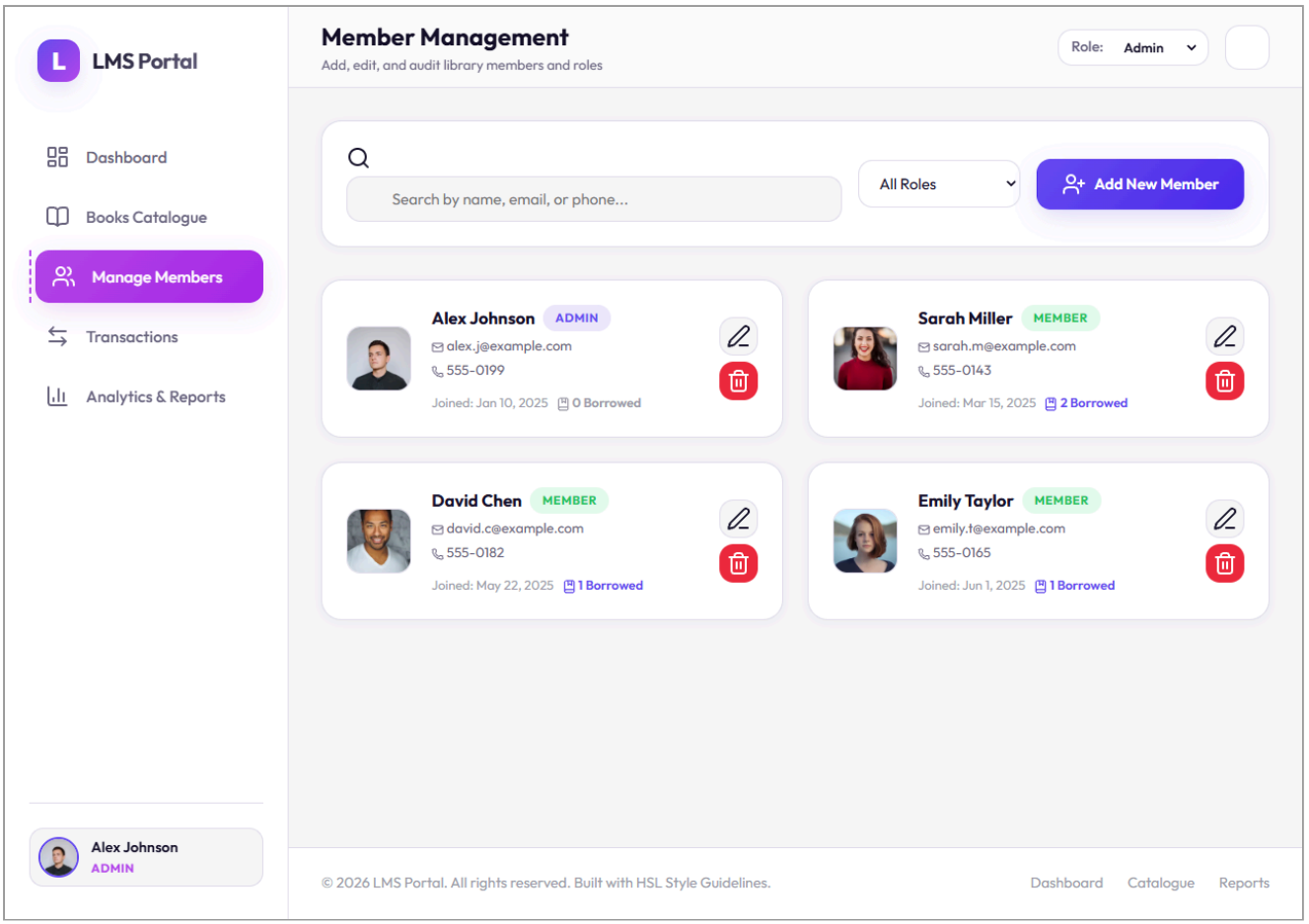
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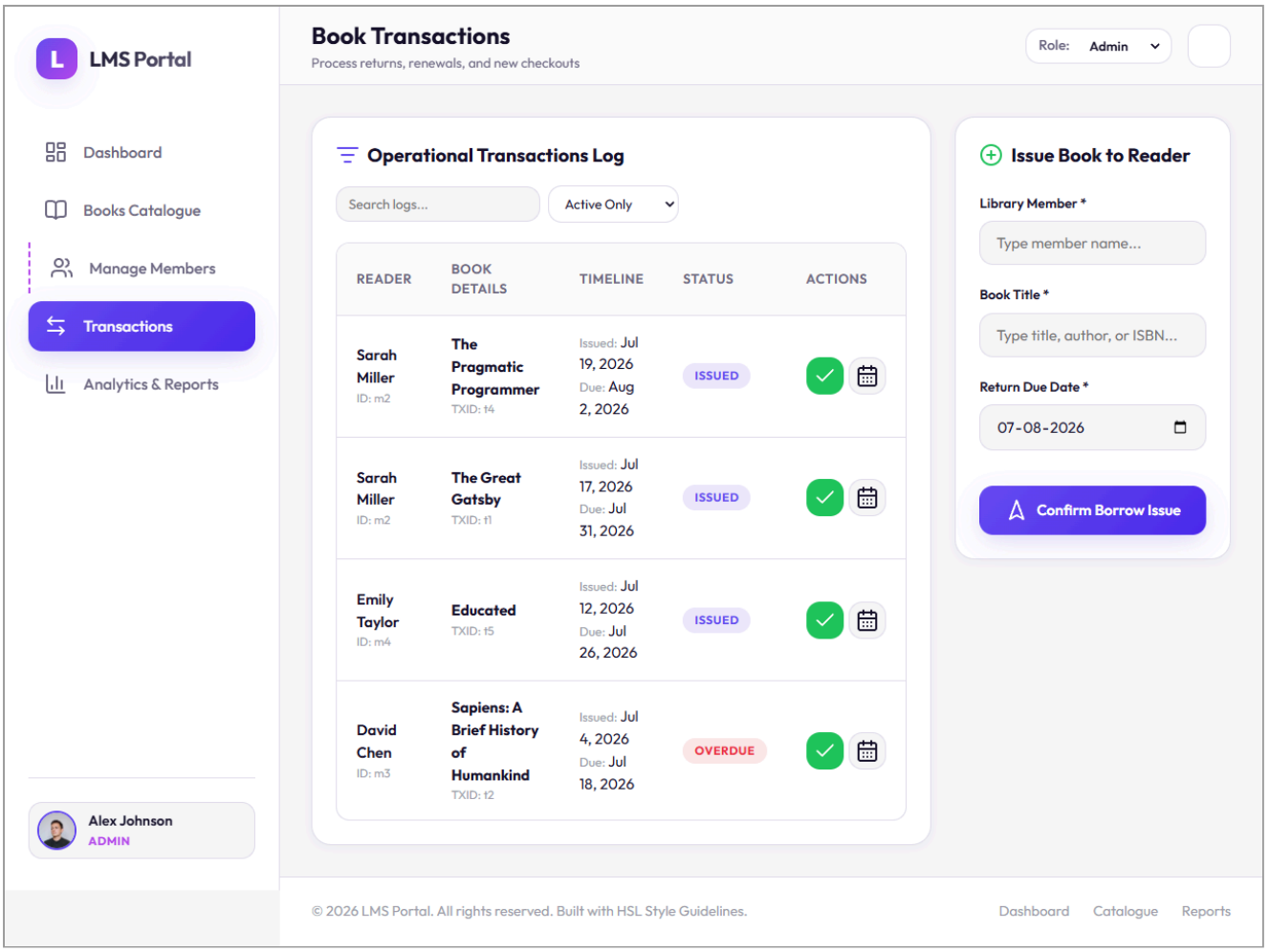
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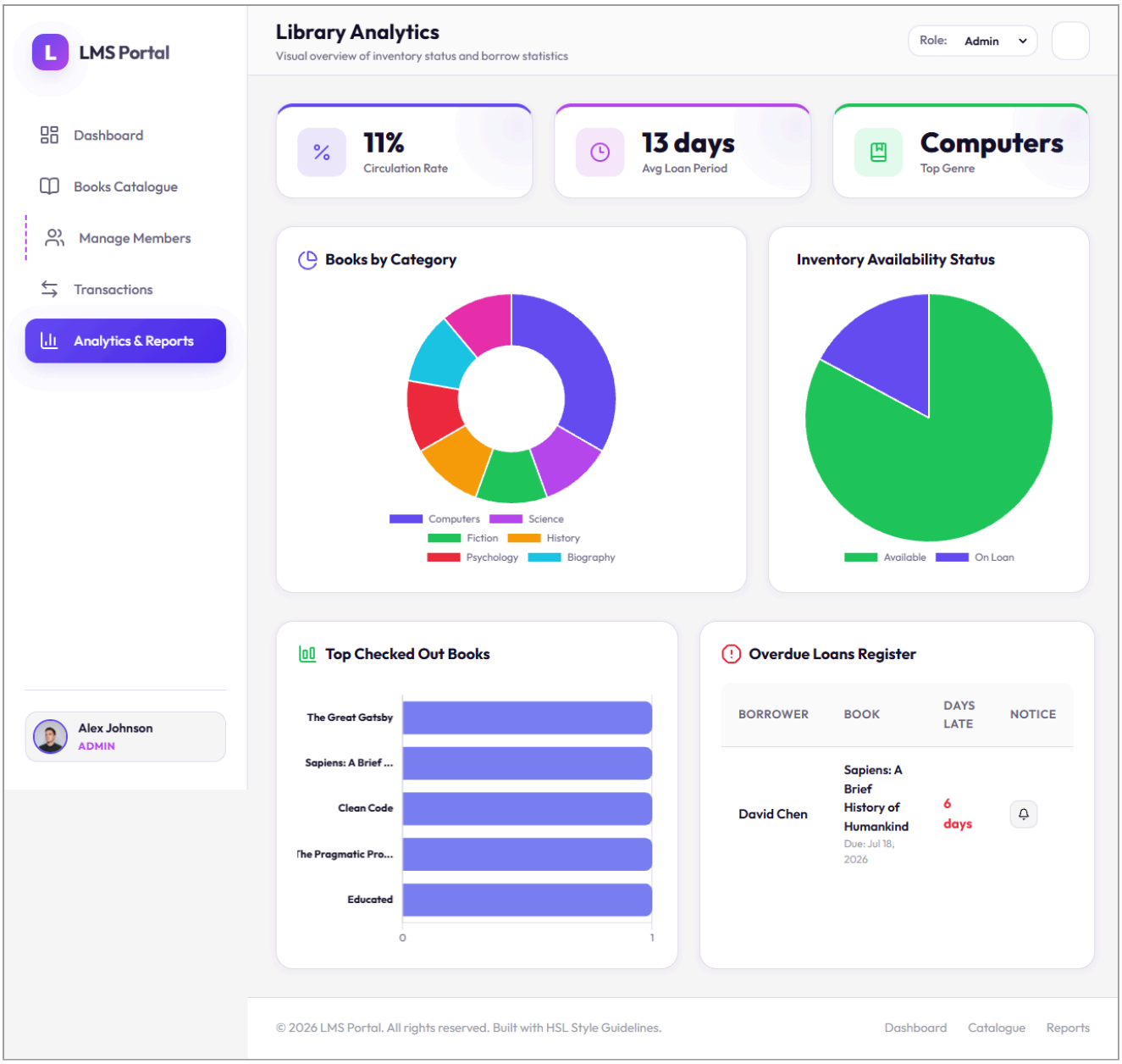
Books Page



Members Page



Transactions Page



Reports Page

7. Edge-Case Testing

The following edge-case scenarios were tested to ensure system robustness:

#	Test Case	Expected Behaviour	Outcome
1	Issuing a book with zero available copies	Issue button is disabled; "Out of Stock" badge shown	Pass
2	Returning a book past the due date	Status automatically set to "Overdue"; fine notice displayed	Pass
3	Deleting a book that has active transactions	Book removed only after all copies are returned	Pass
4	Member accessing Admin-only pages (Members page)	Redirected to Dashboard with an access-denied toast	Pass
5	Searching with an empty or invalid query	All items displayed; no errors thrown	Pass
6	Adding a book with missing required fields	Modal validation prevents submission; field highlighted	Pass
7	Editing a member email to an already-used email	Duplicate email detected; save blocked with warning	Pass
8	Issuing a book to a non-existent member ID	Autocomplete prevents invalid selection	Pass
9	Refreshing page while a modal is open	Modal state reset on reload; no ghost overlays	Pass
10	LocalStorage cleared mid-session	Seed data re-initializes automatically on next load	Pass

8. Conclusion

The WEB-LMS project successfully delivers a complete library management experience entirely within the browser. All core functionalities — book and member management, transaction processing, overdue tracking, and analytical reporting — are implemented and tested thoroughly.

The zero-backend architecture makes deployment instant: simply serve the static files with any HTTP server. The use of LocalStorage eliminates the need for databases while still providing persistent, session-independent data storage.

This project demonstrates proficiency in front-end web development including semantic HTML5, advanced CSS3 with custom properties, ES6+ JavaScript, third-party library integration (Lucide, Chart.js), and browser automation testing with Puppeteer.